ASSIGNMENT 1

PAYMENT CALCULATOR FOR A VILLAGE MILK CO-OPERATIVE

Task 6: Organise the Logic into Functions

Aim: To divide the milk payment calculator into reusable functions.

Functions used:

`rate_per_litre(fat, snf)` calculates the rate per litre using the fat and SNF percentages and returns the rate.

`payout(litres, fat, snf)` calculates the farmer's payment and returns the amount rounded to two decimal places.

`ten_day_total(payments)` adds the payment amounts for the ten-day period and returns the total.

Parameters and return values: Parameters provide the input values to a function. The `return` statement sends the calculated result back to the calling code. Returning a value is different from printing it: a returned value can be stored, reused and tested.

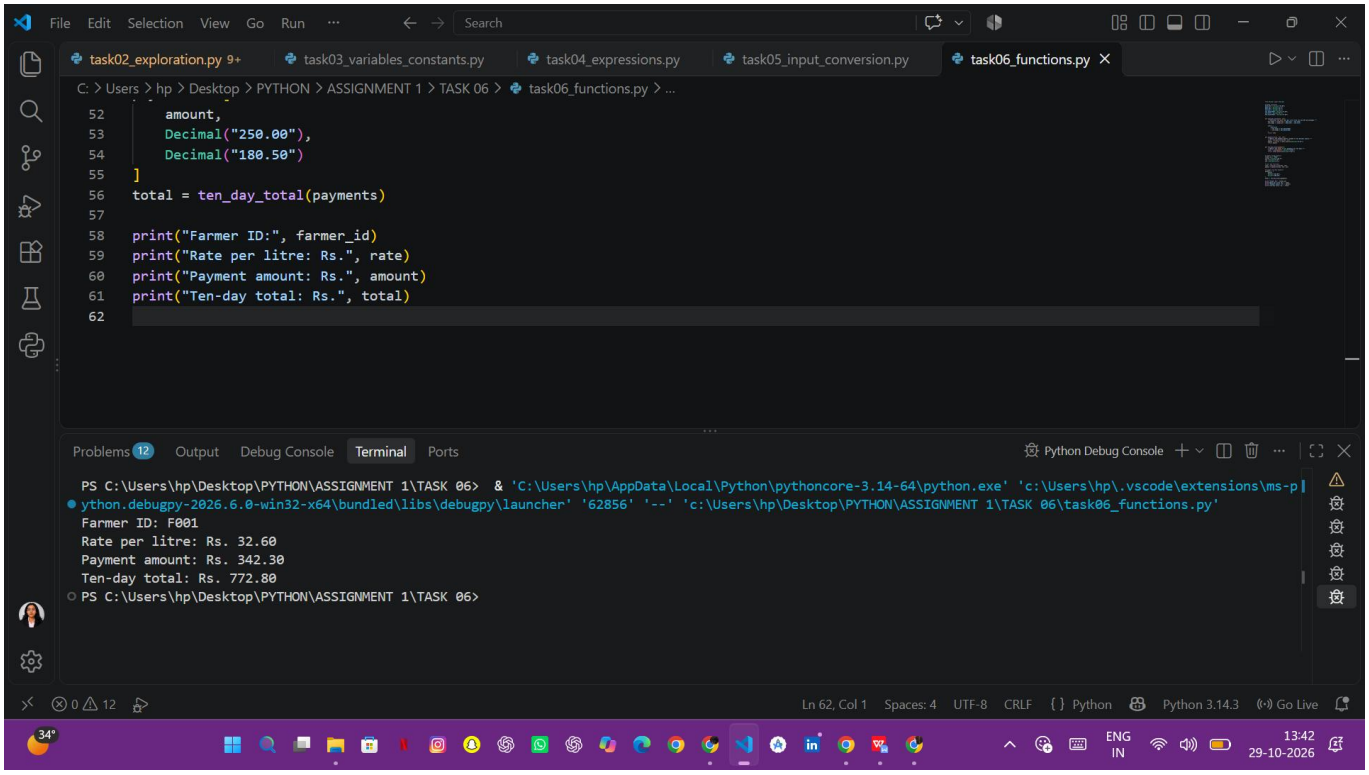
Local and global variables: Variables such as `fat_steps`, `snf_steps`, and `amount` inside functions are local variables. The named rate constants are defined globally and read by the functions. The functions do not modify these constants.

Docstrings: Each function has a short docstring describing its purpose.

Alternative concept: If all calculations were written in one long script, the code would be harder to maintain and reuse. Lambda expressions can represent short expressions but are less suitable for several calculation steps and detailed documentation.

Complexity: Calculating one rate or payment takes O(1) time and O(1) auxiliary space. Calculating the total of n payments takes O(n) time and O(1) additional space, excluding the list that stores the payments.

Conclusion: Functions make the calculator modular, reusable, easier to test and simpler to maintain when the payment rules change.


File Edit Selection View Go Run
Search
task02_exploration.py 9+
task03_variables_constants.py
task04_expressions.py
task05_input_conversion.py
task06_functions.py
C: > Users > hp > Desktop > PYTHON > ASSIGNMENT 1 > TASK 06 > task06_functions.py > ...
amount,
Decimal("250.00"),
Decimal("180.50")
]
total = ten_day_total(payments)
print("Farmer ID:", farmer_id)
print("Rate per litre: Rs.", rate)
print("Payment amount: Rs.", amount)
print("Ten-day total: Rs.", total)
Problems 12
Output
Debug Console
Terminal
Ports
Python Debug Console
PS C:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 06> & 'C:\Users\hp\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62856' '--' 'c:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 06\task06_functions.py'
Farmer ID: F001
Rate per litre: Rs. 32.60
Payment amount: Rs. 342.30
Ten-day total: Rs. 772.80
PS C:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 06>
Ln 62, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.14.3
Go Live
ENG
IN
13:42
29-10-2026